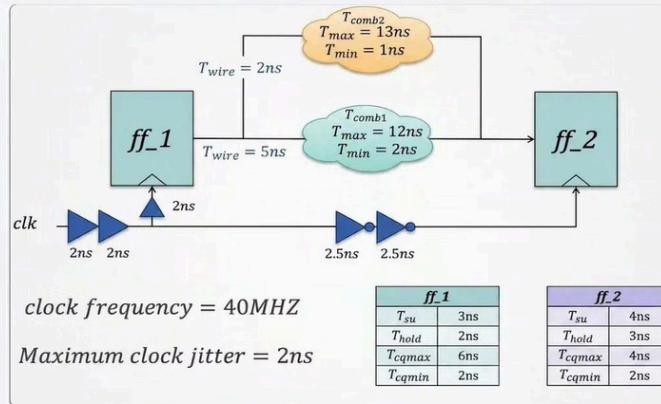


STA problems



- 1- State all the possible timing violation ?
- 2- State the method that used to solve setup violation and hold violation if any ?

STA

Mostafa

52

Copyright © 2026 NTI

Setup values from the diagram

- Clock period: $T = 1/40\text{MHz} = 25\text{ns}$
- Clock jitter: **2ns**
- Clock insertion delay to ff_1: $2+2+2 = 6\text{ns}$
- Clock insertion delay to ff_2: $2+2+2.5+2.5 = 9\text{ns}$
- Clock skew (ff_2 later than ff_1): $9 - 6 = 3\text{ns}$

Two data paths exist from ff_1 to ff_2:

- **Path A** (through Tcomb1): delay = $T_{wire}(5) + T_{comb1} \rightarrow \text{max}=17\text{ns}, \text{min}=7\text{ns}$
- **Path B** (through Tcomb2): delay = $T_{wire}(2) + T_{comb2} \rightarrow \text{max}=15\text{ns}, \text{min}=3\text{ns}$

1) Setup check (uses T_{cq_max} , T_{su} , max path delay, jitter subtracted from margin)

$$\text{Data required} = T_{clk}(ff2) + T - T_{su}(ff2) - \text{Jitter} = 9 + 25 - 4 - 2 = 28\text{ns}$$

2) Hold check (uses T_{cq_min} , T_{hold} , min path delay, jitter subtracted from margin too)

$$\text{Data required (hold)} = T_{clk}(ff2) + T_{hold}(ff2) + \text{Jitter} = 9+3+2 = 14\text{ns}$$

Q1 — possible violations:

- **Setup violation** on the path through **Tcomb1** (the long/slow path), slack = -1ns
- **Hold violation** on the path through **Tcomb2** (the short/fast path), slack = -3ns

The *slowest* path fails setup, the *fastest* path fails hold.

Q2 — fixes:

Setup fix (Tcomb1 path):

- Speed up the logic: resize/upscale cells, reduce logic levels, optimize the combinational block to cut Tcomb1_max
- Reduce wire delay (better buffering/placement/routing)
- Use a faster FF for ff_1 (lower Tcq_max) or ff_2 (lower Tsu)
- Apply **useful skew**: delay the clock to ff_2 further (or advance it to ff_1) since positive skew here already helps setup — push it further in that direction
- Reduce clock jitter/uncertainty if possible
- Last resort: relax the clock frequency (increase T)

Hold fix (Tcomb2 path):

- **Insert delay buffers** into the Tcomb2 data path to increase Tpath_min (this is the standard fix — hold violations are almost always fixed by adding delay, never by changing clock frequency, since hold is frequency-independent)
- Reduce the skew in the opposite direction (bring ff_2's clock arrival closer to ff_1's, i.e., reduce that 3ns skew) — note this trades against the setup fix, so there's a balancing act
- Choose a FF with better Tcq_min/Thold characteristics if available